Marketing Campaign Analysis

Data Cleaning, Statistics, Machine Learning & Business Insights Report

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Executive Summary

This report analyzes a messy, multi-currency, multilingual marketing campaign dataset spanning 8 countries and 6 channels. After a fuzzy-matching cleaning pipeline resolved typos, currency formats, and geo-inconsistencies, 1033 of 1260 rows (82.0%) were retained for analysis. A machine learning model was trained to predict conversion rate from campaign attributes, and concrete budget-reallocation recommendations were derived directly from the cleaned data.

1,033	6.08%	$13,087,827	-0.001
Rows Analyzed	Avg Conversion	Total Budget	Best Model R^2

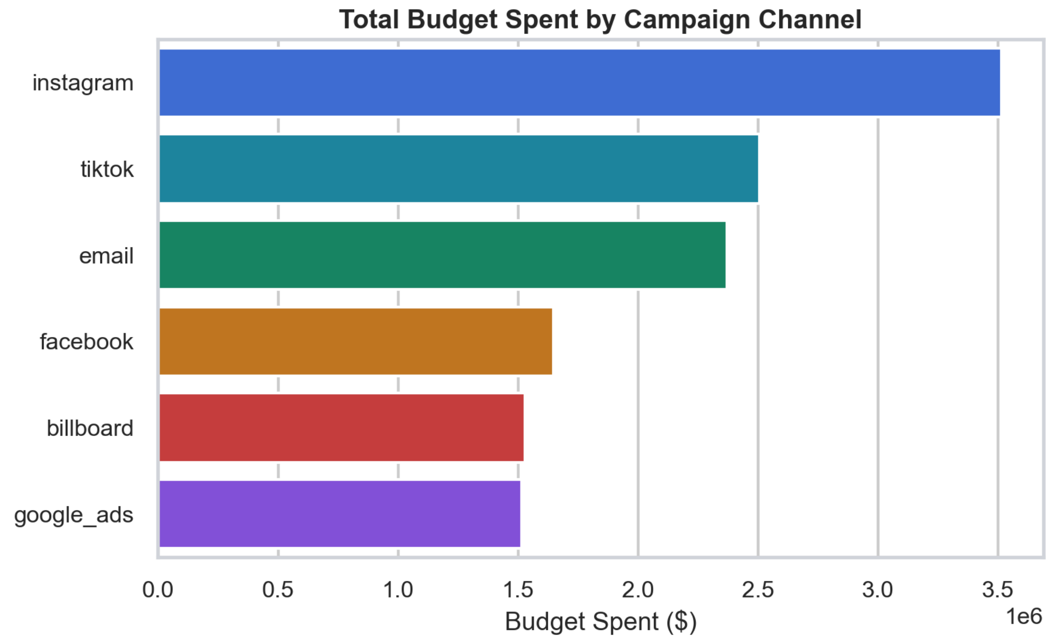
Data Quality & Cleaning Summary

The raw dataset contained 1260 rows with typos, mixed currency symbols (€/$), percent signs, decimal commas, duplicated/garbled category strings, inconsistent date formats, and 1-20% missing values per column. After cleaning, 1033 rows (82.0%) remained. Corrections applied: normalized text casing and accents; fuzzy-matched typo variants of country, city, product, category and channel names against a canonical vocabulary (threshold 0.72 similarity); corrected country values to match known cities; stripped currency/percent symbols and coerced numeric columns; parsed inconsistent date formats to ISO; and imputed remaining gaps via column mean (numeric) or mode (categorical). The 'impressions' column was dropped entirely — its values were too corrupted and inconsistently scaled to trust for analysis.

Rows before cleaning: 1260 → after cleaning: 1033

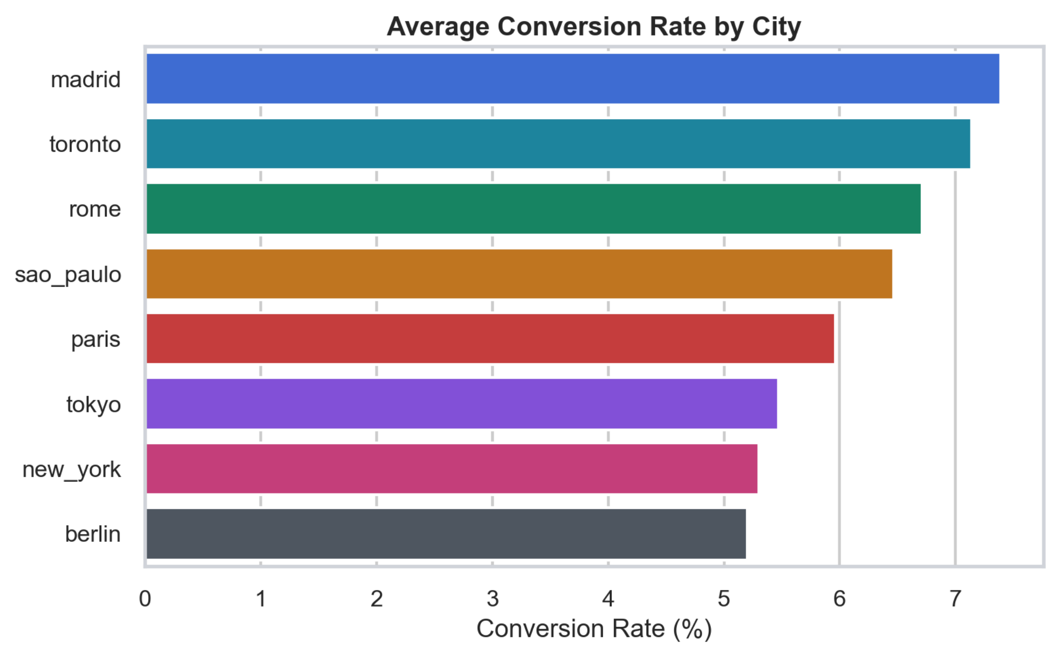
Descriptive Statistics

Budget Spent by Campaign Channel



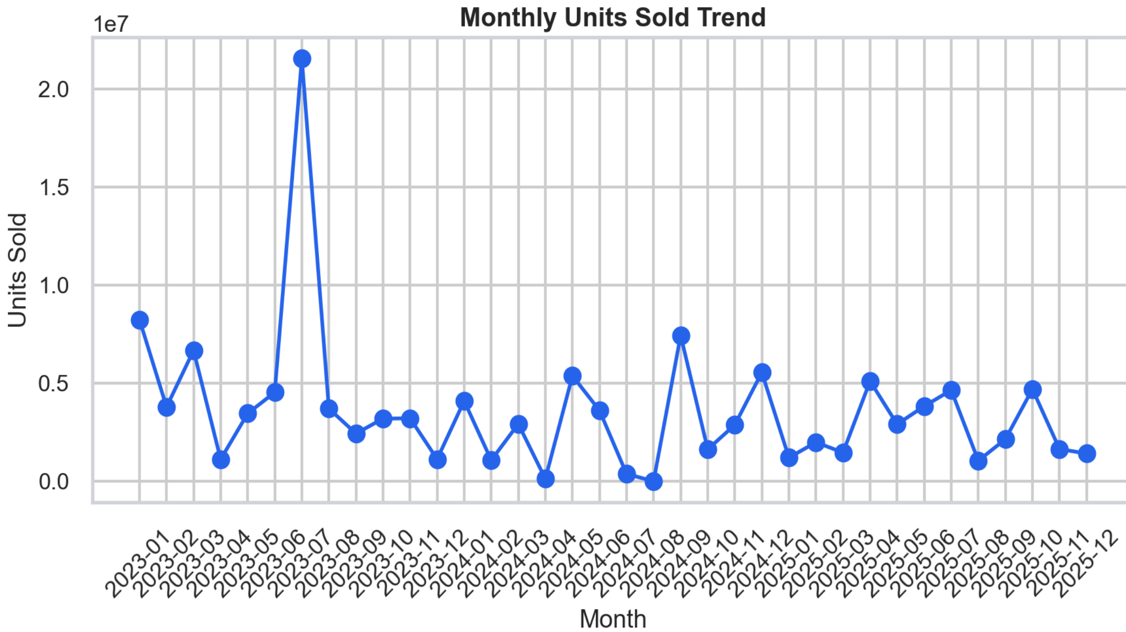
Total budget allocated across each marketing channel.

Conversion Rate by City



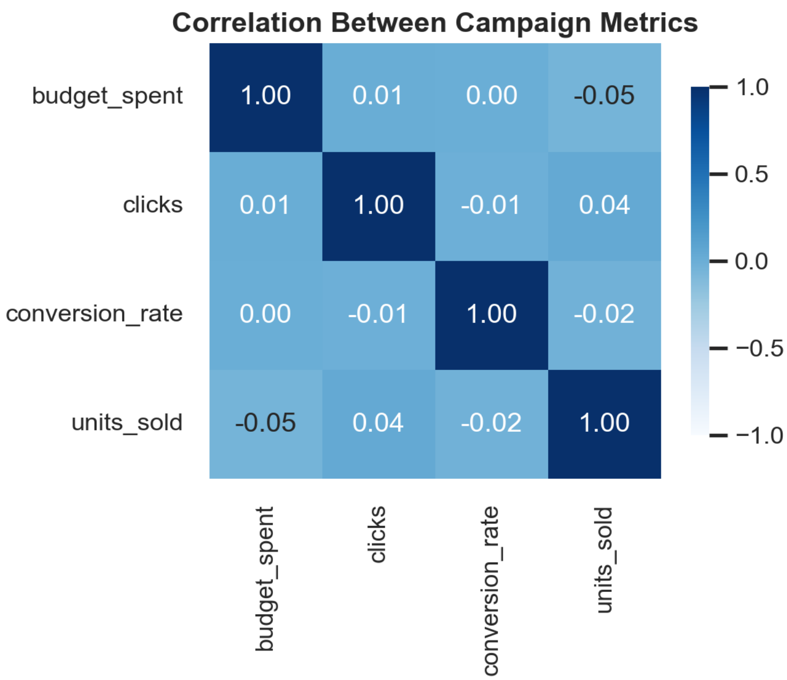
Average conversion rate per city, cleaned and geo-corrected.

Monthly Units Sold Trend



Time trend showing total units sold aggregated by month.

Correlation Between Campaign Metrics



Pearson correlation across budget, clicks, conversion rate and units sold.

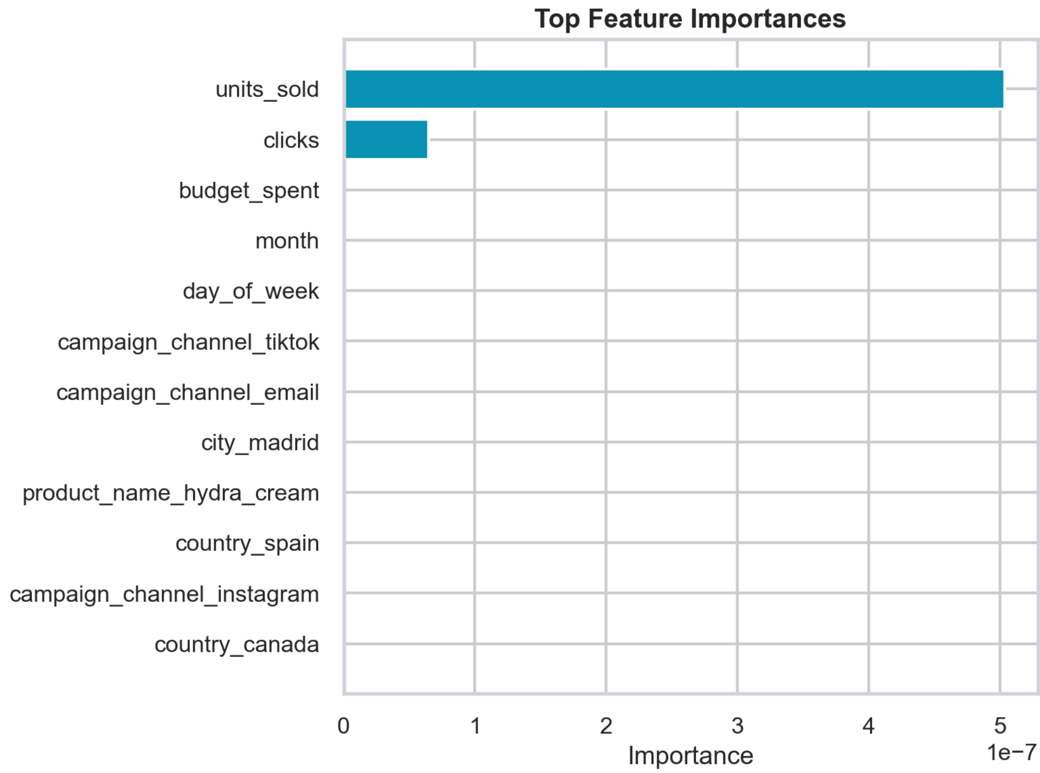
Machine Learning: Conversion Rate Prediction

A regression model was trained to predict conversion_rate from campaign attributes (country, city, product, category, channel, budget, clicks, units sold, and date-derived month/day-of-week). Three candidates were compared via 5-fold cross-validation and a held-out test set: Linear Regression, Random Forest, and Gradient Boosting.

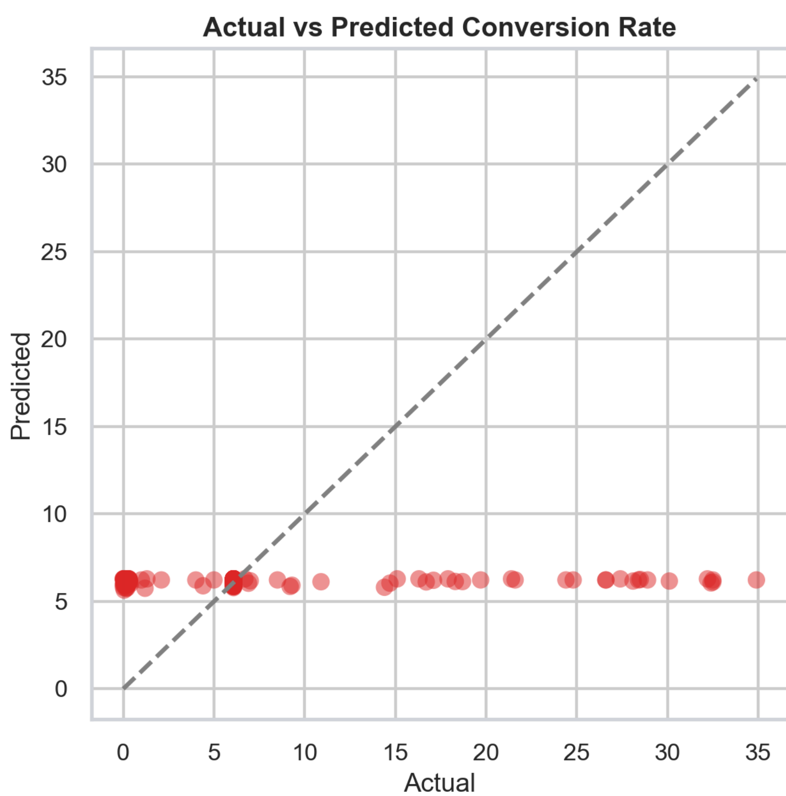
Model	CV R^2 (mean)	Test R^2	Test RMSE
LinearRegression (best)	-0.027	-0.001	7.994
RandomForestRegressor	-0.104	-0.083	8.312
GradientBoostingRegressor	-0.140	-0.089	8.337

Winning model: **LinearRegression** with test R^2 = -0.001 and RMSE = 7.994.

Top Feature Importances



Actual vs Predicted Conversion Rate



Business Insights & Recommendations

Best and worst channel by budget efficiency

'google_ads' delivers 13.826 units sold per $ spent, the highest of all channels, while 'tiktok' trails at 7.445 units per $ — a 1.9x efficiency gap.

Highest converting channel

'instagram' has the highest average conversion rate at 6.55%, versus a dataset average of 6.08%.

Top converting cities

Top 3 cities by average conversion rate: madrid (7.39%), toronto (7.14%), rome (6.71%). Lowest is berlin at 5.20%.

Budget allocation vs. sales output by country

'france' receives the largest budget share ($2,191,132), while 'japan' actually sells the most units (22,123,939). This budget-to-output misalignment suggests reallocating spend toward 'japan' could improve overall ROI.

Best performing product category

'electronics' converts best at 6.68% on average, ahead of 'furniture' at 5.73%.

Strongest numeric relationship

'budget_spent' and 'units_sold' show the strongest correlation in the dataset at r=-0.05.

Recommendations

1. Reallocate incremental budget toward 'google_ads', which returns 13.826 units sold per $ spent versus 7.445 for 'tiktok'.

2. Scale creative testing on 'instagram' campaigns, which already convert at 6.55% on average, above the 6.08% dataset mean.

3. Prioritize localized campaigns in 'madrid', the top-converting city at 7.39%, to compound existing demand.

4. Investigate why 'france' absorbs the largest budget ($2,191,132) yet 'japan' generates more units sold (22,123,939); shifting spend could raise blended ROI.

5. Retire or redesign the lowest-efficiency channel/city combinations identified above before the next budget cycle, and re-test with a smaller pilot spend.

Appendix: Methodology

Cleaning: string normalization, fuzzy canonicalization (difflib SequenceMatcher, threshold 0.72), currency/percent stripping, numeric coercion, date parsing, city-implies-country geo correction, and mean/mode imputation. Statistics: pandas groupby aggregations, pivot tables, and Pearson correlation. Machine learning: scikit-learn ColumnTransformer with one-hot encoding, 80/20 train/test split, 5-fold cross-validation, model selection by test R^2/RMSE. Visualization: Plotly for the interactive dashboard, matplotlib/seaborn for static report charts. Report generation: reportlab platypus.